

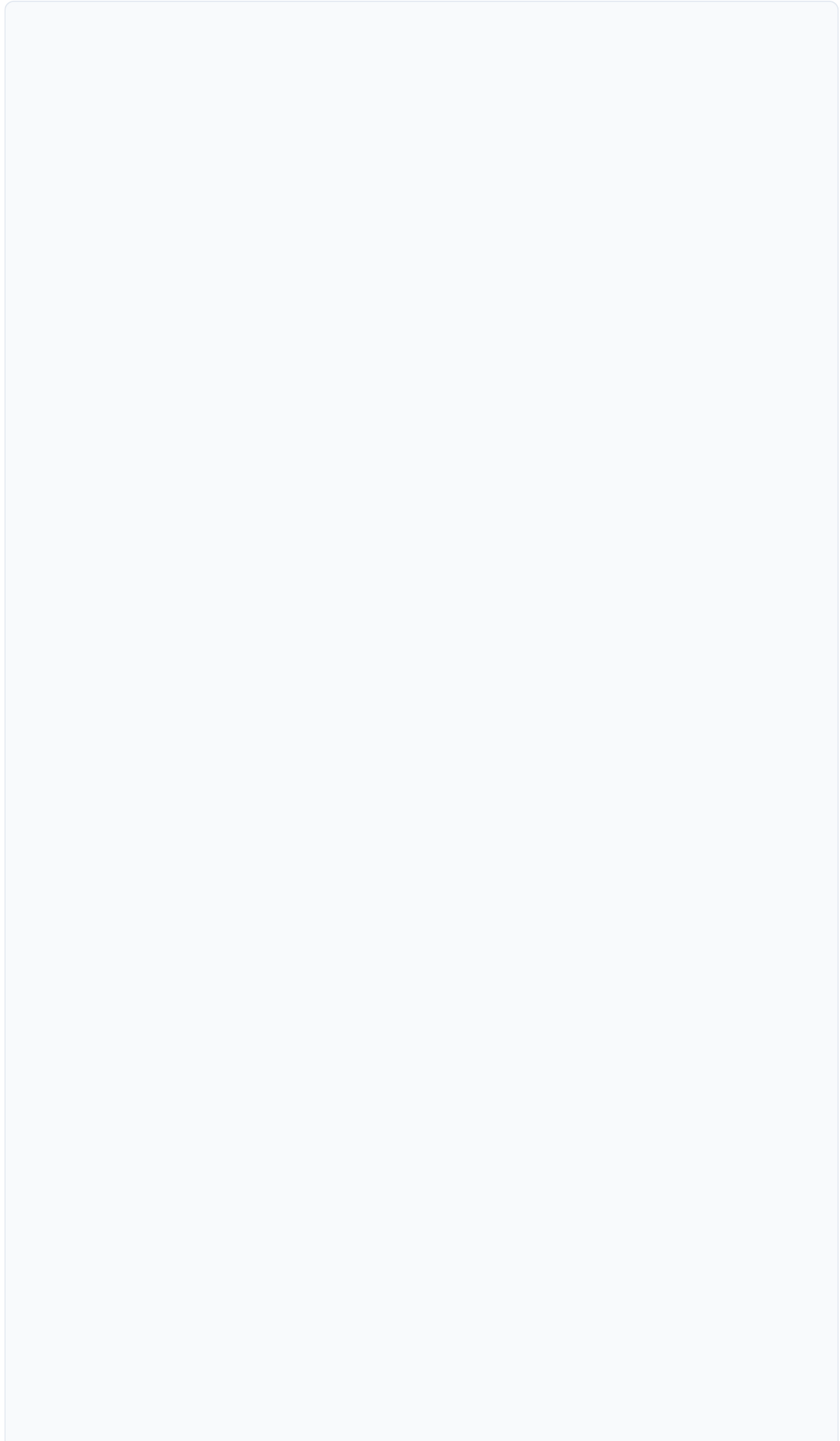
KAPIT-BISIG SYSTEM ARCHITECTURE DOCUMENTATION

Target Beneficiary Flowchart & Lifecycle

End-to-End Analysis: Mobile Proof Submission → Offline Sync → Web Review → Multichannel Verification



System Flowchart Diagram





Step-by-Step Architectural Lifecycle

1

Mobile Event Discovery & Proof Input Mobile App

Residents locate active disaster events on `ResidentHomeDashboardScreen.tsx`. In `ResidentProofRequestScreen.tsx`, they select damage categories (Flood, House Damage, Landslide, Livelihood Loss, etc.) and write damage descriptions.

2

Tamper-Evident Photo Capture & Offline Storage Offline Sync Engine

3 to 5 damage photos are taken via `photoWatermark.ts`, imprinting timestamps, GPS coordinates, and resident IDs into photo pixels. Submissions are saved durably to `ResidentOfflineStore.ts` with status `PENDING_SYNC`.

3

Server Ingestion & Record Creation Backend API

Upon internet reconnect, `ProofSyncCoordinator.ts` posts queued items to `POST /api/beneficiaries/sync/proof-submissions`. The server creates a `ProofSubmission` record (status: *Pending Verification*) and upserts `BeneficiaryEligibility`.

4

Web Dashboard Review & Verification Web Admin

LGU Staff and Admins review pending claims on the `/target-beneficiaries` page (`TargetBeneficiariesPageClient.tsx`). Staff inspect watermarked photos and damage details, submitting an **Approval** or **Revision Request**.

5

Multichannel Notifications & Relief Claiming Notification Engine

The server sends instant FCM push notifications and SMS alerts via `beneficiaryReviewSms.ts`. Approved residents unlock a dynamic beneficiary QR Code for instant scanning at relief distribution centers.

 System Components Summary

Component	Location / File Path	Key Responsibility
Mobile Proof Screen	mobile/components/ResidentProofRequestScreen.tsx	Damage details entry & watermarked photo capture UI.
Offline Store	mobile/services/sync/ResidentOfflineStore.ts	Encrypted local storage for queued offline proof records.
Sync Coordinator	mobile/services/sync/ProofSyncCoordinator.ts	NetInfo network monitoring & automatic sync background worker.
Backend Routes	apps/web/apps/server/routes/beneficiaryRoutes.ts	Express API handlers for proof ingestion & review endpoints.
Beneficiary Service	apps/web/apps/server/services/beneficiaryService.ts	Core approval logic, eligibility updates & notification triggers.
Web Admin UI	apps/web/apps/src/components/beneficiaries/TargetBeneficiariesPageClient.tsx	LGU Staff review table, watermarked gallery viewer & approval modal.